Ocean Engineering 228 (2021) 108695

Contents lists available at ScienceDirect

Ocean Engineering

journal homepage: www.elsevier.com/locate/oceaneng

Estimation of ship routes considering weather and constraints

Wei Du [a], Yanjun Li [a,**], Guolei Zhang [a,*], Chunhui Wang [b], Pan Chen [a], Jipan Qiao [c]

[a] College of Power and Energy Engineering, Harbin Engineering University, Harbin, 150001, Heilongjiang, China
[b] College of Shipbuilding Engineering, Harbin Engineering University, Harbin, 150001, Heilongjiang, China
[c] Shanghai Ship and Shipping Research Institute, Shanghai, 200000, China

ARTICLE INFO

Keywords:
Weather conditions
Constraints
Ship speed loss
Route estimation

ABSTRACT

Taking a T300k VLCC oil tanker as the research object, the ship route estimation model with minimum fuel consumption, CO_2 emissions and shortest ETA as the objective function is established, which considers the weather conditions and constraints of the main engine rated power and safety comprehensively. On the basis of model verification, combined with the weather forecast data and the current maritime market conditions, the estimation indicators were analyzed. The research results show that, prefabricated Route 1 and 2 were the shortest ETA and the minimum fuel consumption routes respectively, although the prefabricated Route 1 saved 36.58 h compared with the prefabricated Route 2, the fuel consumption and CO_2 emissions of the prefabricated Route 2 were reduced by 0.94 t/h and 2.92 t/h, and under the shortest ETA objective function, if keeping the initial speed, the main engine would be limited by rated power when the ship encounters Beaufort number≥4 headwind or bow wind. Accordingly, the ship route estimation method comprehensively considering weather conditions and constraints can analyze the variation rules of the fuel consumption, speed loss, CO_2 emissions and costs of multiple prefabricated routes under different objective functions, which can provide decision support for shipping companies to improve operation economy.

1. Introduction

More than 80% of global trade is carried out by shipping transports, which provides a low-cost and reliable method of international transportation. The International Maritime Organization (IMO, 2018) approved an agreement in 2018 to reduce CO_2 emissions per transport unit by 40% by 2030 and 70% by 2050, compared with 2008. Facing such a severe situation, it is very important for marine energy conservation and emissions reduction and improve the operating efficiency to estimate the fuel consumption, CO_2 emission and economy of prefabricated routes which planned by the captain based on voyage mission, hydrometeorology and ship structure characteristics, in order to select a reasonable route according to the voyage task. At present, the research on ship route estimation mainly focuses on two aspects: one is considering ship safety and crew comfort in ship route design stage (Kanellopoulou, 2019; Yong Bai et al., 2015), the other is ship routing, which aiming at the maximum profit of fleet to reasonably arrange ship routes and ship schedules (Meng et al., 2012). Szlapczynski, R. (2006) used the strategy of analyzing collision avoidance and steering conditions to estimate the safety of routes. Shenping Hu et al. (2017) carried out ship navigation safety risk assessment based on the fuzzy function risk assessment method. Cho, S. C. et al. (1996) proposed a route estimation model based on fleet size and liner routes. Bendall et al. (2001) set an assign tasks model of the container ship branch transportation with the maximum profit goal. Perakis et al. (1991) studied route planning using linear programming and mixed integer programming methods. Jeong et al. (2019) proposed a multiple standard route planning technology, which allows operators to objectively activate routes according to their own intentions. Dimitris Konovessis and Nicolas Bialystocki (2016) estimated the energy consumption of the remaining voyage and proposed decision making and saving methods based on the main factors affecting fuel consumption and speed. Thomas Dickson et al. (2019) proposed a new relative risk assessment model based on fuzzy function on the basis of comprehensive safety assessment method, and this model conducted risk assessment by calculating the probability of occurrence of hazards during ship navigation. Silvia Pennino et al. (2020) used Dijkstra method to detect the optimum route with the highest seakeeping performance of the ship under the

* Corresponding author.
** Corresponding author.
E-mail addresses: Dewey891217@163.com (W. Du), lyjheu@163.com (Y. Li), zhangguolei@hrbeu.edu.cn (G. Zhang).

https://doi.org/10.1016/j.oceaneng.2021.108695
Received 22 July 2020; Received in revised form 29 January 2021; Accepted 31 January 2021
Available online 16 April 2021
0029-8018/© 2021 Elsevier Ltd. All rights reserved.

constraint of maximizing the non-negative cost function, without significantly affecting the voyage duration or the route length.

From the above analysis, it can be seen that the current research on route estimation does not consider the impact of weather conditions, especially rough sea conditions, main engine rated power, safety and other constraints on voyage fuel consumption, CO_2 emissions and economy comprehensively. The activated route is mostly based on the captain's experience and lack of quantitative scientific basis, resulting in the selected route not being the most economical or time-saving route (Huang Daoguang and Wang Lei, 2002). Based on this, under the constraints of the main engine rated power and ship navigation safety, and considering the ship's seakeeping characteristics fully, this paper intends to establish a ship route estimation model. The model can analyze the impact of the difference in weather condition and voyage of multiple prefabricated routes between the departure port and the destination port on important indicators such as ship speed, fuel consumption, estimated time of arrival (ETA), CO_2 emissions, basic costs, etc., and can select the optimum route according to voyage mission, so as to provide a practical way for operating ships to save fuel and improve efficiency (Corbett, James J. et al., 2009).

2. Mathematical model

2.1. Problem description

Taking the VLCC oil tanker as the research object, the ship route estimation method comprehensively considering weather conditions and constraints is established under the constraints of ship speed, the main engine rated power and safety. With shortest ETA, lowest fuel consumption and CO_2 emissions of voyage as objective function respectively, the method can analyze variation rules of ship speed, resistance, fuel consumption, etc. Under the influence of winds and waves, and estimate the key indicators such as voyage distance, ETA, fuel consumption, CO_2 emission and basic cost of different prefabricated route, so as to provide theoretical basis for operating ships to select an appropriate route according to the voyage mission and their concerned targets.

2.2. Calculation of ship navigation resistance

The total ship resistance mainly includes calm water resistance and additional resistance from winds and waves. The additional resistance from winds and waves is the key influencing factors of speed loss, fuel consumption and CO_2 emissions.

2.2.1. Ship resistance in calm water

The calm water resistance $R_{T,i}$ in each sailing leg mainly includes friction resistance $R_{f,i}$ and residual resistance $R_{r,i}$.

$$R_{T,i} = R_{f,i} + R_{r,i} \tag{1}$$

(1) The frictional resistance in each sailing leg is calculated using the flat formula:

$$R_{f,i} = \left(C_{f,i} + \Delta C_f\right) \cdot \frac{1}{2} \rho V_{c,i}^2 S \tag{2}$$

where $C_{f,i}$ and ΔC_f are the coefficient of frictional resistance and roughness subsidy in the i-th sailing leg respectively, ρ is the seawater density (kg/m^3), $V_{c,i}$ is the calm water ship speed in the i-th sailing leg (m/s), and S is the wet surface area (m^2).

The frictional resistance coefficient $C_{f,i}$ is calculated using the Schoenherr formula, and the roughness subsidy coefficient ΔC_f is calculated using the ITTC-1957 formula:

$$C_{f,i} = \frac{0.4631}{(\lg \mathrm{Re}_i)^{2.6}} \tag{3}$$

$$\mathrm{Re}_i = \frac{V_{c,i} \cdot L_{WL}}{\nu} \tag{4}$$

$$\Delta C_f = \left[105 \left(\frac{K_s}{L_{WL}}\right)^{\frac{1}{3}} - 0.64\right] \times 10^{-3} \tag{5}$$

where Re_i is the Reynolds number in the i-th sailing leg, L_{WL} is the designed waterline length (m),ν is the kinematic viscosity (m^2/s), K_S is the hull surface roughness, where $K_S = 150 \times 10^{-6}$ calculated using the ITTC method.

(2) A Lap-Keller graph is used to calculate residual resistance. Through polynomial fitting of the graph, the residual resistance coefficient C_r is obtained by interpolation, and the residual resistance correction coefficient is introduced at the same time. The residual resistance calculation formula in each sailing leg is as follows:

$$R_{r,i} = a \cdot (\xi / 100 + 1) \cdot C_r \cdot \frac{1}{2} \rho V_{c,i}^2 S \tag{6}$$

$$a = A_1 \cdot \sin(B_1 L_{WL} + C_1) + A_2 \cdot \sin(B_2 L_{WL} + C_2) \tag{7}$$

where a is the residual resistance correction coefficient, ξ is the percentage of length-width correction, and A_1,B_1,C_1,A_2,B_2 and C_2 are the parameters obtained by fitting a number of ship model test data.

Finally, we combine (2) and (7) to obtain the total resistance in calm water:

$$R_{T,i} = R_{f,i} + R_{r,i} = \left[C_{f,i} + \Delta C_f + a(\xi / 100 + 1) C_r\right] \cdot \frac{1}{2} \rho V_{c,i}^2 S \tag{8}$$

2.2.2. Ship speed loss in winds and waves

As a consequence of the added resistance due to winds and waves, the ship speed is often reduced from the service speed so called involuntary speed reduction. The Kwon's method can predict the involuntary drop of speed in irregular waves and winds. In order to maintain the constant speed, the main engine is necessary to provide more power under rough sea conditions. Townsin and Kwon (1993), Kwon (2008) proposed a corresponding formula for various sea states:

$$\frac{\Delta V_i}{V_{c,i}} \times 100\% = C_{\beta,i} C_{U,i} C_{F,i} \tag{9}$$

$$\Delta V_i = V_{c,i} - V_{s,i} \tag{10}$$

$$V_{c,i} = Fr_i \cdot \sqrt{L_{pp} \cdot g} \tag{11}$$

where ΔV_i is the i-th sailing leg ship speed difference (m/s), $V_{s,i}$ is the i-th sailing leg ship speed in the selected weather conditions (m/s),$C_{\beta,i}$ is the i-th sailing leg speed direction reduction coefficient, depending on wind direction angle (with respect to the ship's bow, as shown Fig. 1) and the Beaufort number (BN) (as shown Table 1), $C_{U,i}$ is the i-th sailing leg speed reduction coefficient, varying with ship's block coefficient C_B, loading conditions and the i-th sailing leg Froude number Fr_i, $C_{F,i}$ is the i-th sailing leg hull form coefficient (as shown in Table 2), a function of ship type, the Beaufort number and ship displacement ∇(m^3), Fr_i is the i-th sailing leg Froude number, which associated with designed ship operation speed $V_{c,i}$ in calm water conditions, L_{PP} is ship length between perpendiculars (m), g is the acceleration of gravity (m/s^2) (as shown in Table 3).

During the ship's navigation, not only the involuntary speed reduction caused by winds and waves, but also the voluntary speed reduction caused by safety constraints must be considered. In order to ensure

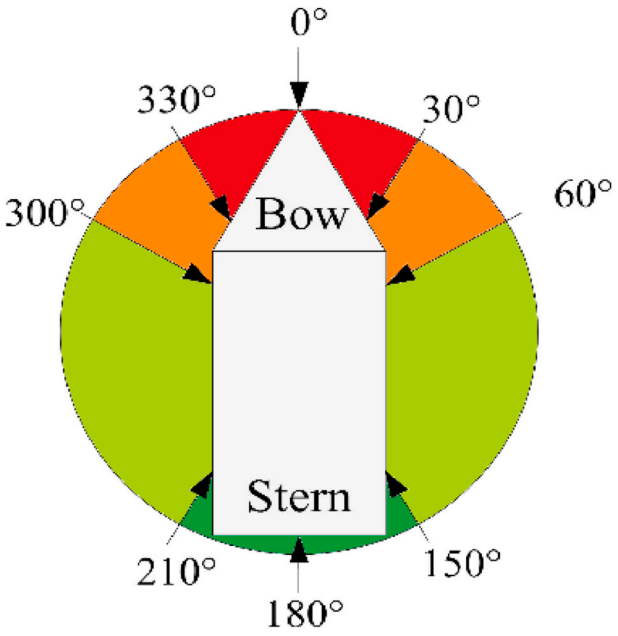


Fig. 1. Weather direction angle.

Table 1
Speed direction reduction coefficient $C_{\beta,i}$ (Kwon, 2008).

Weather direction	Direction angle (with respect to ship's bow) β	Direction reduction coefficient $C_{\beta,i}$
Head sea (irregular waves) and wind	0°~30°	$2C_\beta = 2$
Bow sea (irregular waves) and wind	30°~60°	$2C_\beta = 1.7 - 0.03(BN-4)^2$
Beam sea (irregular waves) and wind	60°~150°	$2C_\beta = 0.9 - 0.06(BN-6)^2$
Following sea (irregular waves) and wind	150°~180°	$2C_\beta = 0.4 - 0.03(BN-8)^2$

Table 2
Speed reduction coefficient $C_{U,i}$ (Kwon, 2008).

Block coefficient $C_{\beta,i}$	Ship loading conditions	Speed reduction coefficient $C_{U,i}$
0.55	Normal	$1.7 - 1.4Fr_i - 7.4(Fr_i)^2$
0.60	Normal	$2.2 - 2.5Fr_i - 9.5(Fr_i)^2$
0.65	Normal	$2.6 - 3.7Fr_i - 11.6(Fr_i)^2$
0.70	Normal	$3.1 - 5.3Fr_i - 12.4(Fr_i)^2$
0.75	loaded or normal	$2.4 - 10.6Fr_i - 9.5(Fr_i)^2$
0.80	loaded or normal	$2.6 - 13.1Fr_i - 15.1(Fr_i)^2$
0.85	loaded or normal	$3.1 - 18.7Fr_i + 28.0(Fr_i)^2$
0.75	Ballast	$2.6 - 12.5Fr_i - 13.5(Fr_i)^2$
0.80	Ballast	$3.0 - 16.3Fr_i - 21.6(Fr_i)^2$
0.85	Ballast	$3.4 - 20.9Fr_i + 31.8(Fr_i)^2$

navigation safety, for higher Beaufort number ($BN \geq 7$), involuntary speed loss is as important as voluntary, and voluntary speed loss needs to be considered at this time. In addition, Prpić-Oršić and Faltinsen (2012) also showed that the voluntary speed loss is more significant than the involuntary speed loss only when significant wave height (Hs) > 3 m. Here, the navigation safety is judged by ship's synchronous rolling wavelength $\lambda_{syn,i}$ under rough sea conditions. When $\lambda_{syn,i}$ is between ship's length LOA and the maximum energy wavelength λ_{max-E} (40 times

Table 3
Ship form coefficient $C_{F,i}$ (Kwon, 2008).

Type of ship	Ship form coefficient $C_{F,i}$
All ships (except for containerships) in loaded conditions	$0.5BN + BN^{6.5}/(2.7\nabla^{2/3})$
All ships (except for containerships) in ballast conditions	$0.7BN + BN^{6.5}/(2.7\nabla^{2/3})$
Containerships in normal loading conditions	$0.5BN + BN^{6.5}/(22\nabla^{2/3})$

Hs), the ship must consider the voluntary speed reduction and sails at a speed less than the critical speed in the winds and waves (Krata and Szlapczynska,2018, Fengwu Wang, 1998).

$$\lambda_i = 1.56\tau_i^2 \tag{12}$$

$$\tau_{e,i} = \frac{\lambda_i}{1.25\sqrt{\lambda_i} + V_{s,i}} \tag{13}$$

$$\lambda_{syn,i} = \left(V_{s,i}\tau_{e,i} + 0.78\tau_{e,i}^2\right) + \sqrt{\left(V_{s,i}\tau_{e,i} + 0.78\tau_{e,i}^2\right)^2 - V_{s,i}^2\tau_{e,i}^2} \tag{14}$$

$$V_{s,i} < V_{\text{limit},i} = V_{c,i}\left[1 - \left(\frac{m}{LBP} + n\right)\right], \quad if \ \lambda_{syn,i} \ is \ between \ LOA \ and \ \lambda_{max-E} \tag{15}$$

where λ_i, τ_i and $\lambda_{syn,i}$ are wavelength(m), wave period(s) and ship's synchronous rolling wavelength(m) in the i-th sailing leg respectively, LBP is Length between perpendiculars(m), m and n are coefficients that depend on the course and Beaufort number respectively, as shown in Table 4.

2.3. Calculation of sailing time

The ship sailing time depends on the voyage and speed. The difference of routes is not only reflected in the difference of voyage, but more importantly, the encounter of different winds and waves having a significant impact on the ship speed. Therefore, it is necessary to divide the route to several sailing legs according to the weather information on the route such as the wind direction angle and BN, as shown in Fig. 1. The total sailing time T_{total} is the sum of the navigation time of each sailing leg:

$$T_{total} = 0.5144\sum_{i=1}^{N}\frac{Dis_i}{V_{s,i}} \tag{16}$$

where N is total number of the sailing legs, Dis_i is voyage of the i-th sailing leg (nm).

2.4. Calculation of fuel consumption and emissions

2.4.1. Calculation of voyage fuel consumption

The total fuel consumption on the voyage consists of two parts: one is the diesel generator fuel consumption F_{GE} related to the sailing time; the other is the main engine fuel consumption F_{ME} which has a great impact on the total navigation fuel consumption.

Table 4
m and n (Aertssen method).

	head sea and wind		bow sea and wind		beam sea and wind		following sea and wind	
BN	m	n	m	n	m	n	m	n
5	9	0.02	7.0	0.02	3.5	0.01	1.0	0
6	13.0	0.06	10.0	0.05	5.0	0.03	2.0	0.01
7	21.0	0.11	14.0	0.08	7.0	0.05	4.0	0.02
8	36.0	0.18	23.0	0.12	10.0	0.07	7.0	0.03

The total fuel consumption of the voyage is as follows:

$$F_{total} = F_{ME} + F_{GE} \tag{17}$$

The diesel generator fuel consumption is as follows:

$$F_{GE} = Q_{GE} \times T_{total} \times 10^{-3} \tag{18}$$

$$Q_{GE} = sfoc_{GE} \times P_{GE} \times 10^{-3} \tag{19}$$

where Q_{GE}, $sfoc_{GE}$ and P_{GE} are fuel consumption per hour (kg/h), fuel consumption rate (g/kW·h) and power (kW) of the diesel generator respectively. According to the ship configuration and the diesel generator load is almost unchanged during the voyage, $sfoc_{GE}$ is 190 g/kW·h and P_{GE} is 800 kW.

The main engine fuel consumption is as follows:

$$F_{ME} = 0.5144 \sum_{i=1}^{N} \frac{Dis_i}{V_{s,i}} Q_{ME,i} \tag{20}$$

$$Q_{ME,i} = sfoc_{ME,i} \times P_{ME,i} \times 10^{-3} \tag{21}$$

where $Q_{ME,i}$, $sfoc_{ME,i}$ and $P_{ME,i}$ are main engine fuel consumption per hour (kg/h), fuel consumption rate (g/kW·h) and power (kW) in the i-th sailing leg.

The fuel consumption rate is related to the type and load of main engine. In this paper, the main engine bench test data of VLCC oil tanker are used to fit the data:

$$sfoc_{ME,i} = 17.277\left(B_{ME,i}\right)^3 + 11.226\left(B_{ME,i}\right)^2 - 47.363\left(B_{ME,i}\right) + 180.54 \tag{22}$$

$$B_{ME,i} = P_{ME,i} / P_{SMCR} \times 100 \tag{23}$$

where $B_{ME,i}$ is the main engine load (%) in the i-th sailing leg, P_{SMCR} is the main engine rated power (kW).

Within the allowable range of main engine rated power and safety, the main engine power in each sailing leg includes the calm water main engine power and the additional main engine power under the influence of weather conditions such as winds and waves.

$$P_{ME,i} = P_{TME,i} + \Delta P_{ME,i} \tag{24}$$

$$P_{TME,i} = \frac{P_{T,i}}{\eta_D \cdot \eta_s} \tag{25}$$

$$P_{T,i} = R_{T,i} \cdot V_{s,i} \tag{26}$$

where $P_{TME,i}$, $\Delta P_{ME,i}$ and $P_{T,i}$ are the calm water main engine power (kW), main engine additional power (kW) and ship calm water resistance power (kW) in the i-th sailing leg respectively, η_D is the propulsion efficiency, which is a function of the main engine power obtained by fitting ship model test data, η_S is the shaft efficiency, which is 0.99 generally.

The ship's additional resistance power and main engine power:

$$\frac{\Delta P_{T,i}}{P_{T,i}} = (n+1)\frac{\Delta V_i}{V_{c,i}} \tag{27}$$

$$\Delta P_{ME,i} = \frac{\Delta P_{T,i}}{\eta_S \cdot \eta_D} \tag{28}$$

where $\Delta P_{T,i}$ is the additional resistance power in the i-th sailing leg (kW),n is an empirical constant related to ship type and loading status, as shown in Table 5.

2.4.2. Calculation of voyage emissions

Ship emissions are closely related to ship fuel consumption, and selecting different prefabricated route has a great impact on ship emissions. For example, although a certain prefabricated route has a short voyage, the fuel consumption is not the least affected by winds and waves etc. So, it is great significance to select an appropriate prefabricated route for ship energy conservation and emissions reduction.

Table 5
Value of n in the Kwon method.

Ship type	Loading condition	n
Oil tanker	Full load	1.19
	Ballast	2.40
Container ship	–	2.16

$$E_{total} = E_{ME} + E_{GE} \tag{29}$$

$$E_{ME} = F_{ME} \cdot C_{F,ME} \tag{30}$$

$$E_{GE} = F_{GE} \cdot C_{F,GE} \tag{31}$$

where E_{total}, E_{ME} and E_{GE} are the total CO_2 emissions (t), main engine CO_2 emissions (t) and diesel generator CO_2 emissions (t) respectively, $C_{F,ME}$ and $C_{F,GE}$ are the main engine CO_2 emission factor (kg (CO_2)/kg (Fuel)) and diesel generator CO_2 emission factor (kg (CO_2)/kg (Fuel)) respectively. The emission factors of various fuels are shown in Table 6 (ISO 8217, 2010).

2.5. Calculation of navigation costs

The costs incurred during a complete voyage are numerous and complexes, which are not easy to count, so the costs incurred during the voyage only consider charter fees, fuel costs and emission costs here (Perakis et al., 1991). The total cost of the voyage is as follows:

$$C_{total} = Pc_{Rental} \cdot T_{total} + Pc_{Fuel} \cdot F_{total} + Pc_{Emi} \cdot E_{total} \tag{32}$$

where Pc_{Rental}, Pc_{Fuel} and Pc_{Emi} are the daily charter fee ($/day), fuel price ($/t) and CO_2 emission fee ($/t) respectively.

2.6. Ship route estimation model

After the ship accepts a new voyage task, the captain and the second mate carefully study the nautical charts, the port of departure and arrival, the hydro-meteorological according to the voyage task and the ship's characteristics, and preplan the route on the premise of safety by virtue of navigation experience. However, the hydro-meteorological conditions of different prefabricated routes lead to the ship resistance increase differently, and the ships will suffer different speed loss, resulting in different indicators such as ETA, fuel consumption, CO_2 emissions and costs. Based on this, considering the constraints of the main engine rated power and safety, and avoiding the subjectivity of the route planning, the objective function and constraints are constructed as follows in order to obtain the shortest ETA or the lowest fuel consumption and CO_2 emissions prefabricated route more accurately.

The shortest ETA objective function:

$$\min T = \min\left(T^1_{total}, T^2_{total}, T^3_{total} \cdots\right) \tag{33}$$

Table 6
CO_2 emission factor for marine fuel (International standard for marine fuel ISO 8217).

Fuel type	Reference standard	Carbon content	CO_2 emission factor
MDO/ MGO	ISO8217, DMX to DMC level	0.875%	3.20600
LFO	ISO8217, RMA to RMD level	0.860%	3.15104
HFO	ISO8217, RME to RMK level	0.850%	3.11440

The minimum fuel consumption and CO_2 emissions objective function:

$$\min F = \min\left(F_{total}^1, F_{total}^2, F_{total}^3 \cdots\right) \tag{34}$$

$$\min E = \min\left(E_{total}^1, E_{total}^2, E_{total}^3 \cdots\right) \tag{35}$$

Constraints:

$$\begin{cases} \lambda_{syn,i} \notin [LOA, \lambda_{\max-E}], \lambda_{\max-E} \approx 40 \cdot Hs_i, i = 1, \cdots, N \\ V_{s,i} - V_{\max} \le 0, i = 1, \cdots, N \\ V_{\min} - V_{s,i} \le 0, i = 1, \cdots, N \\ P_{ME,i} - P_{\max,ME} \le 0, i = 1, \cdots, N \\ ETA \le ETA_{\text{limit}} \end{cases} \tag{36}$$

where Hs_i is the significant wave height in the i-th sailing leg (m), $V_{\max}$ is the maximum speed (knot), $V_{\min}$ is the minimum speed (knot), $P_{ME,i}$ is the main engine power in the i-th sailing leg (kW), $P_{\max,ME}$ is the main engine rated power (kW), ETA_{limit} is the Expected Time of Arrival required by voyage mission, (h).

By solving the constrained multivariable nonlinear objective function, the estimation indicators of voyage, ETA, fuel consumption, costs and CO_2 emissions of each prefabricated route are obtained, and the variation rules of speed, additional resistance power, main engine power and main engine fuel consumption per nautical mile with the weather conditions in each sailing leg of the prefabricated routes are analyzed, and then the effects of weather conditions, especially rough sea conditions on the estimation indicators of ship fuel consumption, emissions and economy are also analyzed.

3. Model verification

The accuracy of ship's calm water resistance model and speed loss model in winds and waves directly affects the accuracy of the route estimation model calculation. Therefore, the data of ship model test and main engine bench test are used to verify the calm water resistance model respectively, and the ship speed loss model in winds and waves is verified by referring to the research results of other scholars.

3.1. Verification of calm water resistance model

Figs. 2 and 3 show the comparison between the simulation results and the ship model test results, and calculation error for calm water resistance at different speeds of the VLCC oil tanker under full load and ballast load respectively. The results show that the maximum errors under full load and ballast are 1.73% and 0.20% respectively, which are not more than 2.0%.

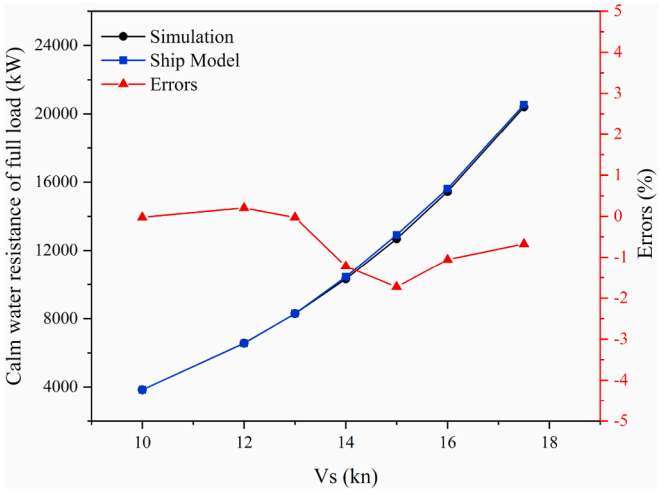


Fig. 2. Calm water resistance power at full load.

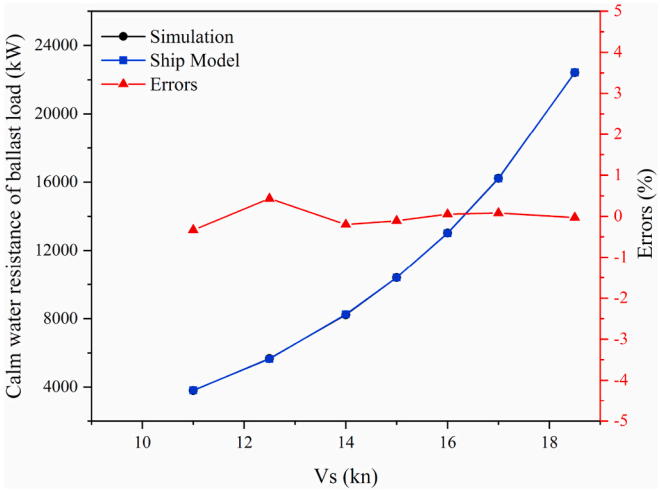


Fig. 3. Calm water resistance power at ballast load.

Figs. 4 and 5 show the comparison between the simulation results and the main engine bench test data provided by MAN Company under full load and ballast respectively (MAN B&W 7G80ME-C9.5, 2017). It can be seen from the figures that the maximum errors under full load and ballast are 1.52% and 0.30% respectively, meeting the calculation requirements.

3.2. Verification of ship speed loss model

Due to the different design and structure parameters of each ship, the influence of winds and waves is also different. It is difficult to obtain the data of ship speed loss caused by winds and waves. Therefore, by referring to the research results of other scholars on ship speed reduction, we verify the characteristics of ship speed loss by using Kwon method (Kwon, 2008) indirectly. Kwon method has been widely used in the study of ship speed loss under different wind and wave conditions, such as ore carriers (Nikolopoulos, L.,2016; Meng et al., 2016), containerships (Second IMO GHG Study, 2009; Shao et al., 2012; M. Kim et al., 2017), and oil tankers (Second IMO GHG Study, 2009; Lu et al., 2015) etc. In addition, the approximation method in Kwon (2008) showed good accuracy in comparison to more extensive calculation methods, and this method was also recommended by the International Maritime Organization (Second IMO GHG Study et al., 2009) to predict the involuntary speed loss of the ship. With the use of the Kwon's method, the prediction of speed loss can be achieved quickly without

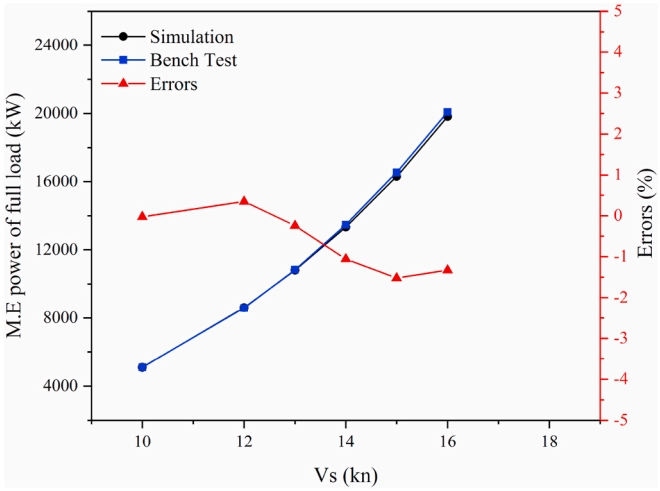


Fig. 4. Main engine power at full load.

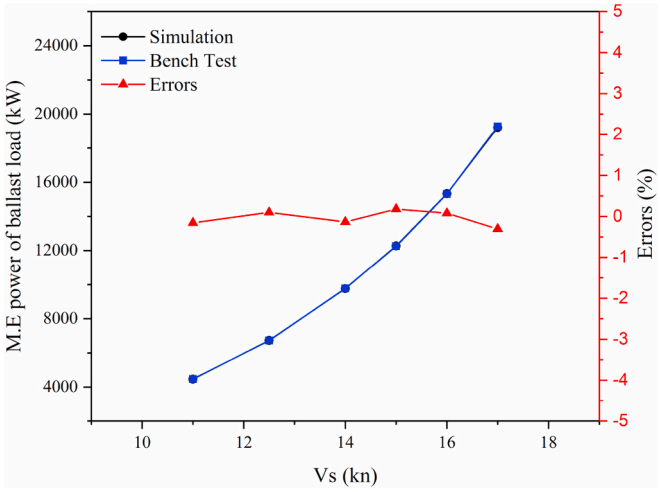


Fig. 5. Main engine power at ballast load.

complex hydrodynamic calculations (Shao, 2013). The Kwon's method (Kwon, 2008) was also adopted to predict added resistance for weather routing in many previous researches such as Larsson and Simonsen (2014), and Shao (2013). It was proved that the Kwon's method (Kwon, 2008) is very fast and reliable on ship operational performance prediction in Lu et al. (2015). Xiaohe Li et al. (2018) also used the same method to verify ship stall in the study of speed optimization considering the influence of irregular winds and waves.

In addition to the influence of winds and waves, under the condition of keeping the main engine power and initial speed unchanged, the variation rule of ship speed loss under the head sea and wind and bow sea and wind when $3 \leq BN \leq 8$ is analyzed respectively, as shown in Fig. 6. As can be seen from the figures, the ship's speed changes nonlinearly with *BN*, and the speed loss under head sea and wind is more than bow sea and wind, which is consistent with the ship's actual sailing speed. Therefore, Kwon method can accurately calculate the ship's stall characteristics under different wind and wave conditions.

4. Route estimation results and analysis

4.1. Shipping market conditions and route information

Considering that the shipping market in different periods is different, fuel oil prices and ship chartering fees also vary greatly. Therefore, according to the recent shipping market information of Clarkson Research Shipping Intelligence Network, the VLCC oil tanker time charter rate is1400 $/d, CO_2 emission fee is 10 $/t, HFO and MDO is 53 $/pmt and 654.5 $/pmt, respectively, and ignoring fuel consumption and emissions when entering and leaving ports and loading and unloading cargo. According to the voyage task and weather forecast data, the ship's different prefabricated routes are estimated. This paper analyzes the basic costs, voyage, ETA, fuel consumption and CO_2 emissions of three prefabricated routes (as shown in Fig. 7) from Tarifa port (Latitude:36.131753°N, Longitude:5.380438°W) to Dos Bocas port (Latitude:19.829503°N, Longitude: 91.92854°W), and the global weather forecast data of August 2019 including u and v components of wind vector, wave height etc.in NetCDF format are downloaded from ECMWF, and the u and v components of wind vector and significant wave height at each turning point are obtained by each turning point information (longitude, latitude and time) interpolation, the true wind direction, *BN* and other information of each turning point are obtained by combining the u and u component vectors, the relative wind direction is calculated by the course and the true wind direction under the assumption that the VLCC ship does not deviate during its navigation. The processed meteorological information is shown in Table 7.

It can be seen from Table 7 and Fig. 7 that the voyage of prefabricated Route 1 and 3 is shorter than that of prefabricated Route 2, but the *BN* in prefabricated Route 2 is very low, and many sailing legs are in the downwind or crosswind, the prefabricated Route 1 and 3 are just the opposite, and even these two prefabricated routes encounter *BN* = 7 headwind or bow skew wind, this sea condition will not only cause the increase of additional resistance and more involuntary speed loss, but also due to the constraints of the main engine rated power and the navigation safety, the ship must slow down.

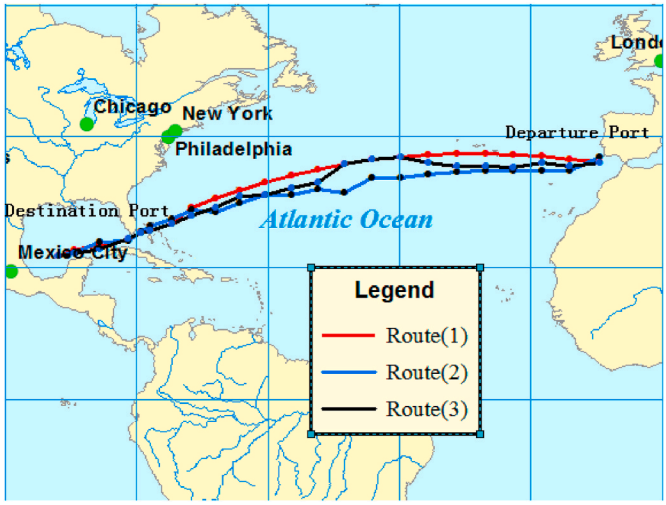


Fig. 7. Information of prefabricated routes.

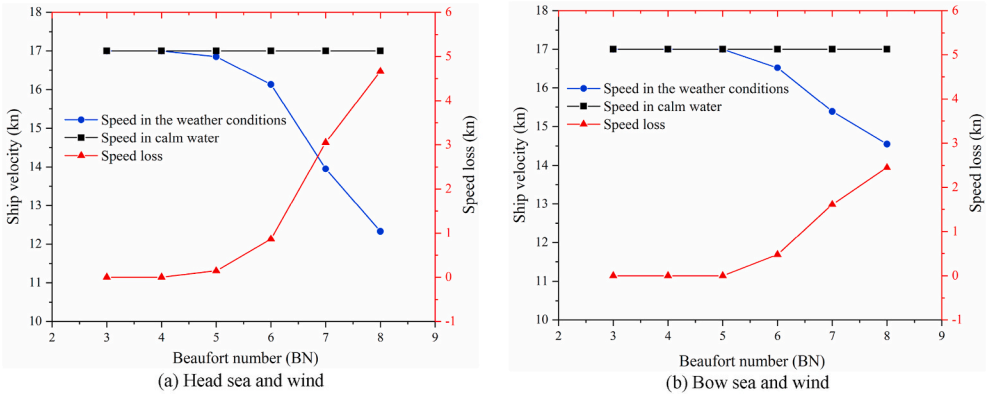


Fig. 6. Variation of ship speed loss in winds and waves.

Table 7
Weather conditions in each sailing leg of prefabricated routes.

Leg	Route (1)				Route (2)				Route (3)			
	Wind direction (°)	*BN*	Wave height (m)	Relative wind direction (°)	Wind direction (°)	*BN*	Wave height (m)	Relative wind direction (°)	Wind direction (°)	*BN*	Wave height (m)	Relative wind direction (°)
1	274.3	5	2.5	25.6	257.3	4	1.2	26.3	260.8	5	3.4	32.3
2	270.3	4	1.3	68.1	268.0	4	1.3	14.1	274.1	5	3.6	36.6
3	250.3	6	3.5	22.6	276.1	4	1.5	5.6	283.5	5	3.6	25.6
4	261.2	6	3.3	6.6	255.2	5	2.2	57.6	232.2	4	1.3	62.6
5	303.4	6	3.2	38.3	240.2	4	1.4	50.8	264.1	5	3.3	33.2
6	262.2	7	3.9	26.1	310.5	3	1.2	31	243.5	5	2.4	56.8
7	283.1	5	2.3	25.6	335.2	5	2.4	28.8	233.2	6	3.3	60.3
8	294.1	5	2.3	24.1	315.1	3	1.3	29.2	193.3	6	3.1	17.6
9	331.5	4	1.2	39.9	347.3	4	1.8	112.6	253.7	6	3.3	31.9
10	232.6	6	3.2	67.4	23.5	5	2.3	37.6	204.4	6	3.4	21.9
11	254.1	7	5.6	52	53.5	5	2.3	27.6	233.7	6	3.3	37.9
12	203.5	6	3.4	12.4	120.3	6	3.2	63.8	213.5	4	1.2	26.9
13	272.7	8	7.4	6.6	136.4	5	2.2	94.5	204.4	4	1.3	52.6
14	102.8	6	3.2	5.4	186.4	4	3.2	72.1	204.1	7	4.5	21.7
15	132.5	5	2.3	0.7	219.6	6	3.1	62.2	173.5	5	2.4	35.6
16	123.6	6	3.2	67.3	232.4	5	2.2	21.5	162.2	5	2.4	26.2
17	112.2	4	1.3	76.1	186.4	4	1.2	92.6	153.3	3	1	25.9
18	113.8	5	2.3	122.3	136.4	4	2.2	94.5	123.3	3	1	21.9
19	102.6	4	1.4	41	162.2	6	3.3	85.7	93.2	5	2.4	29.3
20	174.4	4	1.3	74	154.1	5	2.2	72.3	112.8	4	3.6	33.2
21	184.5	6	3.3	42.4	191.1	5	2.2	122.3	103.3	3	1.1	23.9
22	93	6	3.5	2.6	166.3	5	2.4	27.5	153.3	3	1	26.9
23	84.3	5	2.3	55.7	193.4	6	3.5	62.6	183.5	4	1.2	39.9
24	137.3	4	1.3	16.3	125.6	4	1.3	45.7	104.2	4	1.4	25.5

Note: The weather forecast data from European Centre for Medium-Range Weather Forecasts.

4.2. Route estimation results and discussion

The route estimation calculation program with constraints is developed independently. According to the weather information in Tables 7 and 8 main characteristic parameters of VLCC, the initial speed and the limit ETA are 16 knot and 340 h respectively, and the voyage, ETA, fuel consumption, basic costs and CO_2 emissions of the three prefabricated routes under the shortest ETA, minimum fuel consumption and emissions are calculated respectively, which are shown in Fig. 8.

As can be seen from Fig. 8, prefabricated Route 1 is the shortest ETA route. Compared with prefabricated Route 1 and 2, the ETA saves 36.58 h and 21.08 h, but the fuel consumption increases by 179.00 t and 102.17 t, and the CO_2 emissions also increase by 557.48 t and 318.21 t, respectively. Prefabricated Route 2 is the lowest fuel consumption, the voyage fuel consumption and CO_2 emissions are reduced by 179.00 t, 76.83 t and 557.48 t, 239.27 t compared with prefabricated Route 1 and 3 respectively, however the voyage is 329.23 nm and 386.52 nm longer than the other two routes respectively. It can be seen that, taking the current weather forecast data as an example, if the shortest ETA objective function is chose, ETA per shorten 1 h will result in fuel consumption of prefabricated Route 2 increases by 0.94 t/h and 0.87 t/h compared with the prefabricated Route 2 and 3, the CO_2 emission also increases by 2.92 t/h and 1.78 t/h respectively, and finally the cost increases by 18.27 $/h and 6.44 $/h; if the minimum fuel consumption objective function is chose, although ETA increases 36.58 h and 14.79 h compared with prefabricated Route 1 and 3 respectively, the fuel consumption and CO_2 emission per nautical mile are reduced by 0.05 t/nm, 0.03 t/nm and 0.16 t/nm, 0.10 t/nm respectively, and the cost is also saved by 18.27 $/nm and 11.83 $/nm. This shows that the ship route estimation method comprehensively considering the influence of the voyage and the weather conditions such as winds and waves can estimate multiple ship prefabricated routes, and analyze the differences of estimation indicators among prefabricated routes.

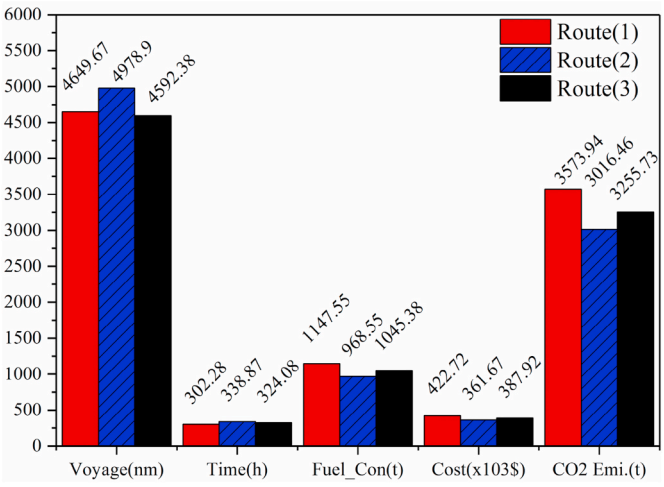


Fig. 8. Estimation indicator of prefabricated routes.

Table 8
Main characteristic parameters of VLCC.

LOA (m)	Beam (m)	Molded depth (m)	Main engine rated power (kW)	Minimum speed (knot)	Maximum speed (knot)	Design speed (knot)
332.95	60	30.5	24720	9	16.5	16

Table 7 shows that due to the differences in the voyage and the relative wind direction, *BN* and *Hs* of each sailing leg, the additional resistance power, main engine power, speed, and fuel consumption per nautical mile are also different under the comprehensive influence of weather conditions and constraints. The variation rules are respectively shown in Figs. 9–12 through calculation and analysis of the route estimation program.

It can be seen from Figs. 9 and 10 that the speed and additional resistance power of prefabricated Route 1 are significantly higher than those of the other two prefabricated routes. If keeping the initial speed 16 knot, the main engine power of almost all sailing legs are higher than the rated power 24720 kW, as shown by the red dotted line in Fig. 11,

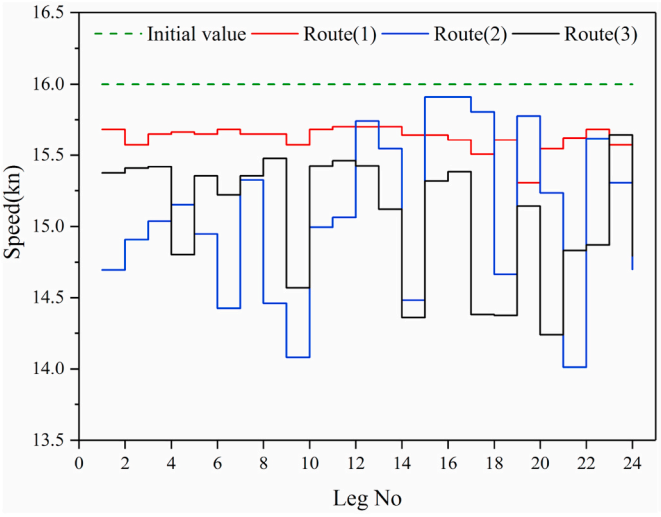


Fig. 9. Speed in each sailing leg.

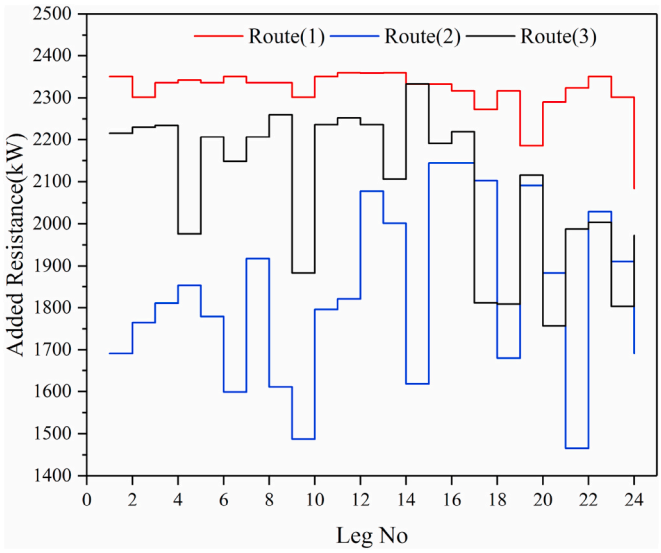


Fig. 10. Additional resistance in each sailing leg.

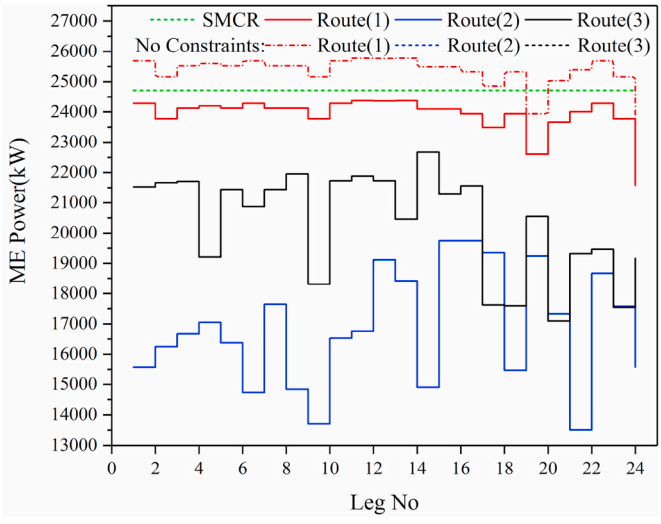


Fig. 11. Main engine power in each sailing leg.

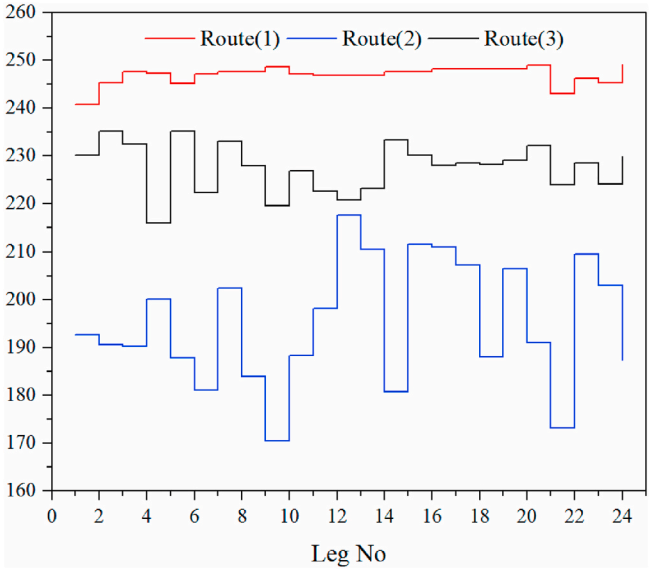


Fig. 12. main engine Fuel consumption per mile of main engine in each sailing leg.

and the maximum value is 25775.76 kW, so under the influence of weather conditions such as winds and waves and the main engine rated power, there are different degrees of speed reduction in each sailing leg, for example, Leg 17 occurs 0.49 knot speed reduction affected by winds and waves and the main engine rated power (as shown in Fig. 11), and because of Leg 19 is near the strait (as shown in Fig. 8), the speed is reduced by 0.69 knot considering navigation safety.

On the contrary to prefabricated Route 1, the speed and additional resistance power of prefabricated Route 2 are less than those of the other two prefabricated routes. At the same time, the main engine power of each sailing leg is less than the main engine rated power, as shown in the blue dotted line in Fig. 11, since the main engine power is not limited, the blue dotted line and the solid line coincide. The main engine power of Leg 15 and 16 are the both largest, which is 19739.26 kW, and the speed and additional resistance power of these two sailing legs are also the largest, which are 15.91 knot and 2144.34 kW respectively. In addition, under the constraint of ETA limit, the speed of Leg 9 is the minimum, which is 14.08 knot, except for considering the safety factors. Although the voyage of prefabricated Route 2 is not the shortest, many sailing legs are in sea conditions such as downwind, cross wind and low significant wave height, which greatly reduces the increase of additional resistance power caused by winds and waves, especially rough sea conditions which resulting in speed loss and fuel consumption.

As shown in Fig. 8 that although the voyage of prefabricated Route 3 is the shortest, the fuel consumption and ETA are not the minimum. Moreover, the sea conditions of prefabricated Route 3 are similar to those of prefabricated Route 1, the ship encounters headwind or cross-wind in multiple sailing legs, which is bound to increase additional resistance, and although the main engine power of each sailing leg does not exceed the rated power (the black dotted line and solid line overlapped in Fig. 11), but the ship encounters rough sea conditions with $BN = 7$, $Hs = 4.5$ m and headwind in sailing Leg 14, even the ship's synchronous rolling wavelength is 99.79 m which is less than the maximum energy wavelength 180 m, the main engine power and additional resistance power are 22671.89 kW and 2333.18 kW respectively, and the ship stalls at 1.64 knot (as shown in Figs. 9 and 10), Hence under the influence of sea conditions such as winds and waves, shortening the voyage may not save fuel and navigation time.

The fuel consumption per nautical mile of each prefabricated route is shown in Fig. 12. Affected by different sea conditions and speeds, the average fuel consumption per nautical mile of prefabricated Route 1, 2

and 3 are 246.86 kg/nm, 195.14 kg/nm and 227.55 kg/nm respectively, and the fuel consumption per nautical mile of prefabricated Route 1 is higher than the other two prefabricated routes, prefabricated Route 3 is between them. These show that the increasing ship speed and rough sea conditions cause different additional resistance, which lead to different main engine power, and ultimately affects the fuel consumption and emissions of the voyage.

5. Conclusions

Taking T300K VLCC oil tanker as the research object, the mathematical model of ship route estimation that comprehensively considering weather conditions and constraints is established, and the calculation program is developed with C language to estimate the voyage, ETA, fuel consumption, basic costs, CO_2 emissions and other indicators of multiple prefabricated routes, so as to provide a method for operating ships to select an appropriate prefabricated route according to voyage task. The main conclusions are as follows:

1. Comparing the simulation results of calm water resistance with ship model test results, the calculation error of calm water resistance under different loading conditions is less than 2%, which verifies the accuracy and applicability of the calm water resistance model. The variation trend of ship speed loss calculated by the route estimation model under the influence of weather conditions is also the same as other research results, which verifies the applicability of the additional resistance simulation module.
2. According to the current weather forecast data, prefabricated Route 1 is the shortest ETA objective function route, which saves 36.58 h and 21.80 h compared with prefabricated Route 1 and 2 respectively, however, due to encounter sea conditions such as headwind, bow wind and high wave height, the average fuel consumption and CO_2 emissions per nautical mile of prefabricated Route 1 are increased by 20.95% and 7.82% compared with prefabricated Route 1 and 2 respectively, resulting in the cost increase by 14.44% and 8.23%.
3. Under the same sea conditions, prefabricated Route 2 is the lowest fuel consumption and CO_2 emissions objective function route, but its voyage is larger than that of prefabricated Route 1 and 3. Because appropriately increasing the navigation distance is to avoid rough sea conditions and take advantage of favorable weather conditions, then multiple sailing legs of prefabricated Route 2 are in the downwind or crosswind, the ship's additional resistance is also reduced, so the average fuel consumption and CO_2 emission per nautical mile are reduced by 26.51% and 16.61%, and the cost is also saved by 16.88% and 7.26% respectively, moreover the increased cost caused by longer navigation time and voyage is far less than the benefits brought by the reduction of fuel consumption and CO2 emissions.
4. Affected by irregular winds and waves, the additional resistance in each sailing leg of the prefabricated routes has increased to varying degrees, the main engine power needs to be adjusted according to sea conditions. Under the shortest ETA objective function, if the ship maintains the initial speed 16 knot, when the ship encounters $BN \geq 4$ headwind or bow wind and $Hs \geq 1.3$ m, the main engine power exceeds the rated power, and the speed is reduced by at least 0.3 knot under the main engine rated power constraint; under the minimum fuel consumption and CO_2 emissions objective function, in order to reduce the ship's fuel consumption and CO_2 emissions, the ship makes full use of favorable weather conditions to reduce speed within the allowable voyage of the voyage ETA limit. This shows that different objective function leads to different choices of prefabricated routes, and due to the comprehensive influence of weather conditions and constraints, ships have different benefits in terms of fuel consumption, CO_2 emissions and navigation time.

Although some research results have been achieved in the prefabricated routes estimation, there are still some shortcomings in the research method of this paper, for example, weather forecast data only considers the influence of wind speed, wind direction and significant wave height, not ocean currents. In the follow-up work, it will be further improved on the basis of current research.

CRediT authorship contribution statement

Wei Du: Validation, Results, Formal analysis, Software, Writing – original draft. **Yanjun Li:** Writing – review & editing, Supervision, Conceptualization, Writing – review & editing. **Guolei Zhang:** Data curation, Validation, Formal analysis. **Chunhui Wang:** Data curation, Writing – review & editing. **Pan Chen:** Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was sponsored by The High-Tech Ship Project "Intelligent Ship 1.0 R & D Special - Ship Comprehensive Energy Efficiency Management System Development" of Ministry of Industry and Information Technology of the People's Republic of China (Grant number: 2016–544), which we gratefully acknowledge. The authors wish to thank the editors and reviewers for their careful, unbiased and constructive suggestions, which lead to this revised manuscript.

References

Bai, Yong, Jin, Wei-Liang, 2015. Chapter 41 - Formal Safety Assessment Applied to Shipping Industry. Marine Structural Design. Elsevier Ltd.
Bendall, H.B., Stent, A.F., 2001. A scheduling model for a high speed containership service: a hub and spoke short-sea application. Marit. Econ. Logist. 3.
Bialystocki, N., Konovessis, D., 2016. On the estimation of ship's fuel consumption and speed curve: a statistical approach. J. Ocean Eng. Sci. 1, 157–166. https://doi.org/10.1016/j.joes.2016.02.001.
Cho, S.C., Perakis, A.N., 1996. Optimal liner fleet routing strategies. volume 4: transport management. Marit. Pol. Manag. 23 (3), 249–259.
Corbett, J.J., Wang, H., Winebrake, J.J., 2009. The effectiveness and costs of speed reductions on emissions from international shipping. Transport. Res. Transport Environ. 14 (8), 593–598.
Dickson, T., Farr, R., Sear, D., Blake, J.I.R., 2019. Uncertainty in marine weather routing. Appl. Ocean Res. 88.
Fengwu Wang, 1998. Discussion on critical speed of ships sailing in wind and waves. World Shipping 21 (2), 11–12.
Hu, S., Fang, Q., Xia, H., Xi, Y., 2017. Formal safety assessment based on relative risks model in ship navigation. Reliab. Eng. Syst. Saf. 92 (3), 369–377.
Huang, Daoguang, Wang, Lei, 2002. How to do the route design well. Navigation technology 17–18, 000 (001).
IMO, 2014. International maritime organization. http://www.imo.org/en/About/Pages/Default.aspx. (Accessed 16 July 2017).
Jeong, M.G., Lee, E.B., Lee, M., Jung, J.Y., 2019. Multi-criteria route planning with risk contour map for smart navigation. Ocean. Eng. 172 (JAN.15), 72–85.
Kanellopoulou, A., Kytariolou, A., Papanikolaou, A., Shigunov, V., Zaraphonitis, G., 2019. Parametric ship design and optimisation of cargo vessels for efficiency and safe operation in adverse weather conditions. J. Mar. Sci. Technol.
Krata, P., Szlapczynska, J., 2018. Ship weather routing optimization with dynamic constraints based on reliable synchronous roll prediction. Ocean. Eng. 150, 124–137.
Kwon, Y.J., 2008. Speed Loss Due to Added Resistance in Wind and Waves. Naval Architect, pp. 14–16.
MAN B&W 7G80ME-C9 5, 2020. MAN energy & systems. https://marine.man-es.com. (Accessed 16 June 2017).
Meng, Q., Wang, T., S, 2012. Short-term liner ship fleet planning with container transshipment and uncertain container shipment demand. Eur. J. Oper. Res. 223 (1), 96–105.
Meng, X., Yuan, Z., 2016. A minimum fuel consumption speed model considering effect of irregular wind and wave. J. Shanghai Marit. Univ. 37 (1), 19–24.
Nikolopoulos, L., Boulougouris, E., Khorasanchi, M., 2016. Applications of Holistic Ship Theory for the Simulation Driven Optimization of the Design and Operation of Large Bulk Carriers. Rina Energy Effic, Ships.
Pennino, S., Gaglione, S., Innac, A., Piscopo, V., Scamardella, A., 2020. Development of a new ship adaptive weather routing model based on seakeeping analysis and optimization. J. Mar. Sci. Eng. 8 (4), 270. https://doi.org/10.3390/jmse8040270.

Perakis, A.N., Jaramillo, D.I., 1991. Fleet deployment optimization for liner shipping Part 1. Background, problem formulation and solution approaches. Marit. Pol. Manag. 18 (3), 183–200.
Perakis, A.N., Jaramillo, D.I., 1991a. Fleet deployment optimization for liner shipping i. background, problem formulation and solution approaches. Marit. Pol. Manag. 24 (3), 183–192.
Prpic-Orsic, J., Faltinsen, O.M., 2012. Estimation of ship speed loss and associated CO_2 emissions in a seaway. Ocean. Eng. 44 (Apr), 1–10.
Second IMO GHG Study, Buhaug, Ø., Corbett, J.J., Endresen, Ø., Eyring, V., Faber, J., Hanayama, S., Lee, D.S., Lee, D., Lindstad, H., Markowska, A.Z., Mjelde, A., Nelissen, D., Nilsen, J., Pålsson, C., Winebrake, J.J., Wu, W., Yoshida, K., 2009. International Maritime Organization (IMO) London, UK. April 2009.
Shao, Wei, 2013. Development of an intelligent tool for energy efficient and low environment impact shipping. Ajr American Journal of Roentgenology 201 (2), W322–W325.
Shao, W., Zhou, P., Thong, S.K., 2012. Development of a novel forward dynamic programming method for weather routing. J. Mar. Sci. Technol. 17 (2), 239–251.
Szlapczynski, R., 2006. A new method of ship routing on raster grids, with turn penalties and collision avoidance. J. Navig. 59 (1), 27–42.
Townsin, R.L., Kwon, Y.J., 1993. Estimating the Influence of Weather on Ship Performance. Wind Pressure.
Xiaohe, L., Baozhi, S., Qianbo, Z., Yanjun, L., Zhenyu, S., Wei, D., et al., 2018. Model of speed ptimization of oil tanker with irregular winds and waves for given route. Ocean. Eng. 164 (SEP.15), 628–639.